Unbounded coefficient oscillation after unit translation

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Abstract

We give an explicit recursive family of polynomials with positive nonincreasing integer coefficients whose translates by one have unbounded coefficient oscillation. Oscillation means the number of sign changes in the list of adjacent coefficient differences after zeros are deleted. At recursion step r, the degree is $8 \cdot 2^r - 5$ and the oscillation count is at least $2r + 1$. The proof preserves an alternating subsequence of strict differences by an integer scaling and appends two new signs beyond the previous degree.

1 Statement and context

All coefficient sequences are ordered by increasing powers of x. For a polynomial $B(x) = \sum_{j=0}^{D} b_j x^j$, let $\nu(B)$ be the number of sign changes in

$$(b_1 - b_0, b_2 - b_1, \ldots, b_D - b_{D-1})$$

after all zero entries have been deleted. An empty list or a list with one nonzero entry has zero sign changes. No coefficients outside $0, \ldots, D$ are included in this definition. A sequence is *unimodal* if it weakly increases to some index and then weakly decreases; the index may be an endpoint. Thus every nonincreasing sequence is unimodal. Oscillation is a count of changes of direction, not a count of peaks.

Theorem 1. *For every finite real number $R \geq 0$, there is a polynomial $A(x) = \sum_{j=0}^{d} a_j x^j$ such that*

$$a_0 \geq a_1 \geq \cdots \geq a_d \geq 1, \qquad a_j \in \mathbb{Z}, \qquad \nu(A(x+1)) \geq R.$$

More precisely, the recursive family constructed below satisfies

$$\deg A_r = 8 \cdot 2^r - 5, \qquad \nu(A_r(x+1)) \geq 2r + 1 \quad (r \in \mathbb{Z}_{\geq 0}). \tag{1}$$

The public transcription of a 2000 report by Alvarez, Amadis and Rosales [1] poses the unbounded-oscillation question for unit translates of unimodal polynomials. It already contains a positive, nonincreasing integer-coefficient example whose translate has two direction changes. The point here is the uniform recursive construction, not the existence of a first nonunimodal example. The report's authors, title and year are also recorded in Ek's bibliography [2].

The direction of monotonicity matters. Boros and Moll [3] prove unimodality after unit translation for nondecreasing nonnegative coefficients. Alvarez et al. [4] extend this to positive integer shifts, and Wang and Yeh [5] to all positive real shifts. Those input hypotheses differ from the nonincreasing class used here.

2 Two elementary observations

Lemma 1. *Let s be a finite list of real numbers. If s has an ordered subsequence of L nonzero entries with alternating signs, then its nonzero entries have at least $L-1$ sign changes.*

Proof. Deleting zeros preserves the selected entries and their order. Between each consecutive pair of selected entries, the signs begin and end differently, so there is at least one sign change in that interval. These $L-1$ intervals contain disjoint sets of adjacent positions in the zero-deleted list. Their sign changes can therefore be counted separately. □

For $M \in \mathbb{Z}_{\geq 0}$, put

$$S_M(x) = \sum_{k=0}^{M} x^k. \tag{2}$$

The binomial theorem and Pascal's identity give, for $0 \leq j \leq M$,

$$\begin{aligned}[x^j]S_M(x+1) &= \sum_{k=j}^{M} \binom{k}{j} \\ &= \sum_{k=j}^{M} \left(\binom{k+1}{j+1} - \binom{k}{j+1} \right) = \binom{M+1}{j+1}. \end{aligned} \tag{3}$$

Here $\binom{j}{j+1} = 0$. Write $h_j = \binom{M+1}{j+1}$ for these coefficients. Each h_j lies between 0 and the sum 2^{M+1} of its binomial row. Consequently

$$|h_{j+1} - h_j| \leq 2^{M+1} \qquad (0 \leq j < M). \tag{4}$$

All sums in this argument are finite.

3 The recursive construction

Set $A_0 = S_3$. Once A_r has degree d_r, define

$$M_r = 2d_r + 5, \qquad C_r = 2^{M_r+1} + 1, \qquad A_{r+1} = C_r A_r + S_{M_r}. \tag{5}$$

We prove simultaneously that A_r has positive nonincreasing integer coefficients and that, writing $B_r = A_r(x+1)$, its strict adjacent differences contain an ordered alternating subsequence of length $2r+2$, beginning positive and ending negative.

Proof of Theorem 1. For $r = 0$, the coefficients of A_0 are $(1,1,1,1)$, and those of B_0 are $(4,6,4,1)$. Its adjacent differences are $(2,-2,-3)$; select the first two. This proves the invariant at $r = 0$, with $d_0 = 3$.

Suppose the invariant holds for r. Abbreviate $d = d_r$, $M = M_r$, $C = C_r$, and $A' = A_{r+1}$. If $(a_0, \ldots, a_d)$ is the coefficient list of A_r, the list for A' is

$$(Ca_0 + 1, \ldots, Ca_d + 1, 1, \ldots, 1). \tag{6}$$

It is positive, integral and nonincreasing: the old segment retains its order, its last entry exceeds 1, and the remaining entries are all 1. The leading coefficient is 1, so $\deg A' = M$.

Write $B_r = \sum_{j=0}^{d} b_j x^j$ and $D_j = b_{j+1} - b_j$ for $0 \le j < d$. Let

$$i_0 < i_1 < \cdots < i_{2r+1} \le d - 1$$

be the selected indices supplied by the induction hypothesis. Their signs are $+, -, +, -, \ldots, +, -$. For the following coefficient calculation only, put $b_j = 0$ when $j > d$. Equation (3) gives the coefficients $b'_j = Cb_j + h_j$ of $B' = A'(x+1)$ for $0 \le j \le M$. Thus at each old selected index,

$$D'_j := b'_{j+1} - b'_j = CD_j + (h_{j+1} - h_j).$$

The old coefficients b_j are integers, because $A_r \in \mathbb{Z}[x]$. Every selected D_j is therefore a nonzero integer. By (4),

$$|CD_j| \ge C > 2^{M+1} \ge |h_{j+1} - h_j|. \tag{7}$$

Hence D'_j has the same strict sign as D_j at every selected position. This is where the integral lower bound on the old strict differences is used.

Two later positions provide new signs. At $j = d + 1$, both old coefficients are zero. Since $M + 1 = 2d + 6$,

$$\begin{aligned} D'_{d+1} &= \binom{2d+6}{d+3} - \binom{2d+6}{d+2} \\ &= \frac{1}{d+3}\binom{2d+6}{d+2} > 0, \end{aligned} \tag{8}$$

where the ratio of the first binomial coefficient to the second is $(d+4)/(d+3) > 1$. At $j = M - 1$, both old coefficients again vanish, and

$$D'_{M-1} = \binom{M+1}{M+1} - \binom{M+1}{M} = 1 - (M+1) = -M < 0. \tag{9}$$

The indices occur in the required order:

$$i_{2r+1} \le d - 1 < d + 1 < M - 1 = 2d + 4. \tag{10}$$

Appending $d + 1$ and $M - 1$ therefore extends the selected signs by $+, -$, to a sequence of length $2r + 4$. Both are actual adjacent differences within the new degree M; no artificial endpoint differences are counted. This completes the induction.

Lemma 1 now gives $\nu(B_r) \ge 2r + 1$. The exact degree follows from the leading coefficient in (6):

$$d_0 = 3, \qquad d_{r+1} = 2d_r + 5, \qquad d_r + 5 = 8 \cdot 2^r.$$

Thus (1) holds. Finally, for any finite $R \ge 0$, choose

$$r = \max\left\{0, \left\lceil \frac{R-1}{2} \right\rceil\right\}. \tag{11}$$

Then $2r + 1 \ge R$, and $A = A_r$ has every required property. □

4 Scope and source provenance

The result gives a lower bound on oscillation, not an exact count for every r or a minimum possible degree. The input monotonicity is weak and has an endpoint mode. The statement uses the fixed shift 1 and does not assert a simultaneous conclusion for all positive real shifts. The proof is symbolic and self-contained; finite examples are not premises for its unbounded conclusion.

The report mirror [1] displays the date 10 August 2000, the oscillation question in Section 3 on printed page 99, and a definition in terms of changes of direction. It does not explicitly specify how zero differences are treated. Our zero-deletion convention is stated above. An institutional original or page image of that report, and its complete subsequent resolution history, have not been authenticated here. Ek's reference [2] corroborates the bibliographic identity, not that history. Accordingly, no claim of first discovery, first nonunimodal example, or worldwide priority is made.

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References

[1] J. Alvarez, M. Amadis and L. Rosales, *Unimodality and Log-Concavity of Polynomials*, Summer Institute in Mathematics for Undergraduates (SIMU), technical report, 2000. Public transcription dated 10 August 2000, Section 3, printed p. 99; institutional original not authenticated here. `https://www.academia.edu/121303677/Unimodality_and_Log_Concavity_of_Polynomials`.

[2] B. Ek, *q-Binomials and related symmetric unimodal polynomials*, author manuscript, 16 April 2018. Reference [AAR00], p. 32. `https://sites.math.rutgers.edu/~bte14/Articles/Gnk/Gnk.pdf`.

[3] G. Boros and V. H. Moll, A criterion for unimodality, *Electron. J. Combin.* **6** (1999), R10. `https://doi.org/10.37236/1442`.

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[5] Y. Wang and Y.-N. Yeh, Proof of a conjecture on unimodality, *European J. Combin.* **26** (2005), 617–627. Official author version: `https://arxiv.org/abs/0809.1586v1`.